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Social network density predicts partisan animosity

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Abstract

People's social networks are fragmenting and partisan animosity is rising. Three studies highlight the connection between social network density and partisan animosity. Americans with denser networks (where more social contacts know each other) appear to often harbor less animosity toward political opponents. One explanation for this association is feelings of unity: dense social networks feel more unified, and people project these feelings of connection and common identity onto distant fellow Americans. Study 1 supports this idea with a representative sample of Americans: social network density predicts feelings of local unity, which predicts feelings of national unity, which predicts warmth toward and willingness to interact with political opponents. Study 2 attempts to manipulate perceived social network density and finds a "causal cascade" of indirect effects culminating in lower partisan animosity, suggesting that more involved interventions may reduce partisan animosity. Study 3 uses nationwide county-level data from the Social Capital Atlas and Project Implicit to reveal that communities with higher social network density exhibit lower partisan animosity. The results suggest that considering the overarching structures of social networks is crucial to understanding partisan animosity, and that fostering social cohesion could help mend political divides.

Introduction

Partisan animosity in the United States is at record levels (Iyengar et al. 2019; Boxell et al. 2022), stifling productive political discussion and splitting communities down party lines (Iyengar et al. 2019; Jones 2001; Mason 2015). While scholars have identified several interpersonal components of partisan animosity (Mason 2018; Ahler 2014; Enders and Armaly 2019; Westfall et al. 2015; Lees and Cikara 2020), broader social forces may also amplify divides, including the shifting fabric of American social networks. Prior work highlights how partisan animosity can fracture social ties (Chen and Rohla 2018), but the reverse is also worth considering: the dissolution of social networks may intensify partisan animosity. Here we explore the relationship between social network density and feelings of partisan animosity.

Social networks

Each of us is embedded within social networks, which consist of ties (relationships) between nodes (people). People vary in their number, nature, and strength of social ties, but even people with the same quantity, quality, and strength of relationships can have different levels of *density*. Social network density is a measure of whether or how well the people in a social network know each other (formally, the number of social ties within a network divided by the overall possible number of ties). A friend group in which everyone knows everyone else is very dense, whereas a group of commuters on a train who do *not* know each other is less dense.

Community social networks have grown sparser (i.e., less dense) over time. Through the majority of human history, networks were extremely dense with most people living and dying in the same small, closely-knit community they were born into (Thomas and Mark 2013), but now many people instead live in sprawling urban centers surrounded by strangers. Adding to urbanization, the decline in community network density is further driven by people increasingly connecting with others through social media platforms and digital technology (Butler and Matook 2015).

The hollowing out of community networks is revealed by several indicators: over a quarter of people in the United States now live alone, fewer adults participate in local social groups, and fewer Americans know or spend time with their neighbors (Bartolini et al. 2013; Hampton et al. 2009; Putnam 2000). This network sparseness has harmful effects. Loneliness has surged in recent decades, with its risks rivaling those of smoking, obesity, and inactivity (Holt-Lunstad et al. 2010; Wilson and Moulton 2010). Social support has also diminished, with Americans in 2004 reporting 30% fewer confidants than in 1985 (McPherson et al. 2006). We suggest that the reduced density of social networks—and their resulting reduced perception of *unified* social networks—might also correspond with higher partisan animosity.

Density and perceptions of unity

Density is an objective social network measure, but people cannot always “see” their entire social networks. Instead, people mostly sense their local relational environment (Bearman et al. 2004), making their *perceptions* of their social networks especially consequential. We suggest that perceiving yourself to be in a dense social network confers a set of interpersonal benefits.

Feeling like you belong to a dense network helps provide feelings of social support (Lee et al. 2018; Wellman and Frank 2001; Huang et al. 2019). Perceptions of collective emotional investment in dense networks cultivate trust among members, helping to foster healthy relationships (Moolenaar and Sleegers 2010; Igarashi et al. 2008; Buskens 1998). Similarly, at the dyadic level, studies on romantic couples have demonstrated that people who feel a sense of fusion with their partners are more secure in their relationships and likely to address conflicts with positive coping mechanisms (Walsh and Neff 2018).

Dense social networks foster perceptions of unity, creating a feeling that the group coheres together as a single entity, like a sports team, intentional community, or platoon. This sense of unity encourages productive conflict resolution. People in sparse networks can sever social ties without impacting their other relationships, but disputes in dense networks affect the entire group (Yu et al. 2020). In a dense network, avoiding one person means avoiding everyone, so people are motivated to work out their differences.

This dynamic might account for the lower rate of relational conflict in dense networks (Curşeu et al. 2012).

The feelings of unity that social network density provide may extend beyond people's individual social groups to the nation. Past work suggests that feelings of national unity mitigate partisan animosity, because they transcend party boundaries and evoke a superordinate identity (Levendusky 2018). But it may also be the case that feeling unified with one's proximate others can create a sense of national unity.

On the surface, one might assume the opposite relationship between network density and feelings of national unity. After all, if you're deeply connected with your local community, why bother connecting to dissimilar others? But research shows that people extrapolate their local environment to interpret broader contexts (Dickie et al. 1997; Barsalou and Wiemer-Hastings 2005; Lakoff et al. 1999). Cognitive science reveals that our understanding of abstract concepts is grounded in more proximate basic experiences. For instance, people understand social connections based on physically embodied sensations like warmth (Lakoff et al. 1999). Similarly, perceptions of broad social groups—like political parties—are grounded in personal experiences with concrete people (Thagard 2012; Barsalou 1999; Entwisle et al. 2007). When people feel more grounded within unified local communities—via dense social networks—they may see the nation more as a unified whole.

Previous work on attachment to close others supports the idea that perceptions of unity with one's friends and family can generalize outward, fostering closeness toward more distant social groups. Feeling securely attached to parents and romantic partners predicts greater openness, decreased bias, and more inclusive attitudes beyond one's immediate social circle (Mikulincer and Shaver 2022). This enhanced sense of security derived from supportive relationships strengthens trust in others' good intentions and facilitates peaceful approaches to resolving conflicts between groups (Boag and Carnelley 2016). The community unity from dense social networks could facilitate greater feelings of social cohesion, extending even to those with different political views.

We propose that people with dense social networks may project that sense of unity onto the nation at large through a process we call **projected unity**, which may be associated with reduced partisan animosity. A connection between social network density and unity may explain the results from one agent-based model where dense social networks seemed to produce less polarization over time than sparse ones (Mittereder et al. 2021). A stronger sense of unity within one's immediate network may create ripple effects that shape broader societal outcomes, including partisan animosity. Here, we systematically test the proposed model in Fig. 1, exploring whether feelings of unity within dense networks are associated with amplified feelings of national unity and reduced partisan animosity.

Pilot studies

In the initial exploratory phase of our research (Stebbins 2001), we first tested this model in two pilot studies using samples of convenience. Given that the relationship between social network density and partisan animosity had not been explored in previous research, we began with a small sample of 200 participants (see Supplementary Materials). Supporting our predictions, we found that participants with denser social networks were warmer and more open towards political opponents. We then conducted

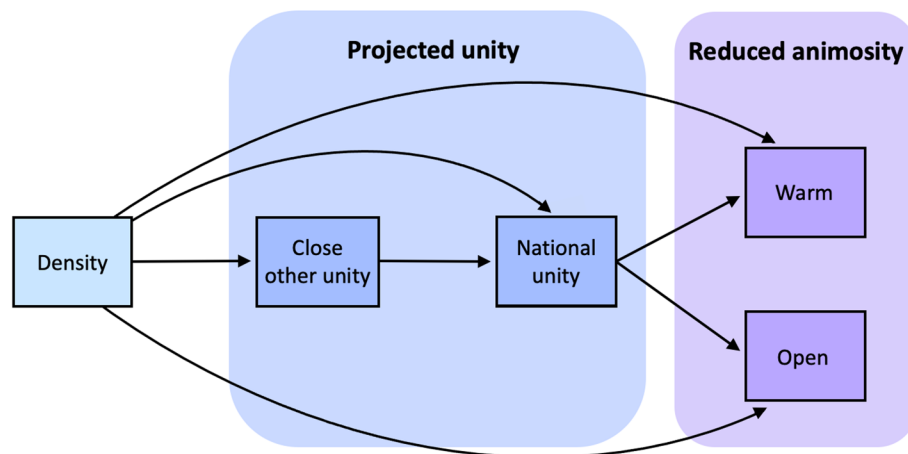


Fig. 1 Full proposed model. Social network density is associated with unity with close others, unity with close others is associated with national unity (the process of projected unity), and national unity is associated with warmth and openness towards political opponents (reduced partisan animosity)

an identical follow-up pilot study with a second convenience sample (see Supplementary Materials) to determine whether our findings would be likely to replicate in future larger studies. In our second convenience sample pilot, participants who lived in *communities* with denser social networks showed less partisan animosity, but the total relationships between *participant*-level social network density and partisan animosity were not significant. To better understand the discrepancy between these two pilot studies, we followed current recommendations (Curran and Hussong 2009; Goh et al. 2016) and conducted an internal meta-analysis testing our model with the combined pilot data. We found that social network density was directly associated with openness towards political opponents, and indirectly associated with warmth towards political opponents through projected unity. We therefore decided to investigate the relationships between social network density, projected unity, and partisan animosity in the following series of high powered pre-registered studies.

Present studies

In a representative sample of the United States population (Study 1), we tested whether people with dense social networks project feelings of unity with their close others onto the nation, and whether this projected unity in turn predicted warmth and openness towards political opponents. We next tested the effects of a potential intervention to decrease partisan animosity, exploring whether prompting people to focus on the denser parts of their social network increased warmth and openness towards political opponents through projected unity (Study 2). Finally, we examined postal code and county-level data on community density and partisan animosity to see whether regions in the United States with more densely connected social networks showed less partisan animosity (Study 3).

Study 1: US representative survey

Among a representative survey of Americans, we investigated whether people with denser social networks—that is, more reported interconnections among their closest friends and family—showed less partisan animosity. We measured social network density by asking participants to report the social relations among the seven people closest

to them, measured their feelings of unity with their close others and distant Americans, and assessed partisan animosity by measuring their warmth and openness towards political opponents. We predicted that participants with denser social networks (when their close connections were also connected with each other) would feel a greater sense of unity with their close others that they would project onto distant Americans, and that this would lead them to feel warmer and more open towards political opponents.

Method

Preregistration

This study is preregistered at <https://osf.io/fpz2v/>.

Participants

We recruited 612 participants representative of the United States population based on gender, ethnicity, age, and political party ($M_{\text{age}} = 45.34$, $SD_{\text{age}} = 16.31$) from a recruitment service called Forthright that specializes in conducting representative surveys. Of these, 72 did not pass an attention check and/or manipulation check, leaving a remaining 540 participants. 257 of the participants were men, 272 were women, 9 were non-binary, and 2 did not specify their gender. 18.6% of participants were Hispanic or Latino/a/x. 71.4% were White, 9.3% were Black, 6.9% were Asian, 1.0% were Native American or Alaskan Native, 7.1% were multiracial, and 4.3% did not specify their race. 42.8% of participants were most closely aligned with the Democratic party, and 57.2% of participants were most closely aligned with the Republican party.

Procedure

We first measured social network density by asking participants to describe the relationships among the seven people interpersonally closest to them. Participants were then told that they would be paired in a conversation with another participant who was from the opposite political party and disagreed with their views on gun control and immigration. To make this conversation setup seem more real, participants were asked to provide a screen name and were shown a waiting screen while they were “matched” with another participant. Participants were told that before speaking with this fellow participant, they would be asked a few questions about their expectations on how the interaction would unfold. After this setup, we measured feelings of unity with close and distant others. We measured partisan animosity prior to the unity questions to ensure that participants were most attentive and engaged for our primary outcome. Our aim was to enhance ecological validity by creating the expectation of a real interaction with a political opponent. Although no actual pairing occurred, animosity measures were collected before this was revealed, helping to preserve the credibility and intended effect of the setup.

Main measures

Social network density To measure social network density, we first asked participants to list the seven people interpersonally closest to them. We then asked them to report the interpersonal closeness among these seven people on a 6-pt scale ranging from complete strangers to extremely close. This was a symmetric measure of closeness between each pair (rather than a perceived closeness rating for each individual person). Density was calculated as the sum of the strengths of the relationships between each pair in the

network divided by the total number of pairs. This method is a commonly used approach to calculate social network density (Broda et al. 2023).

Unity with close others To assess unity with close others, we adapted items from the Identification with all Humanity scale, and the pictorial and text identity fusion measures. We measured feelings of unity separately for both family and friends. We tested the new unity measure in a pilot survey examining the relationship between projected unity and warmth (see Supplementary Materials). A confirmatory factor analysis indicated that the items used to assess close other unity formed separate family and friend factors.

- To what degree do you think of the following groups of people as “family?”
- How close do you feel to each of the following groups?
- How much do you identify with (that is, feel a part of, feel love toward, have concern for) each of the following?
- How often do you use the word “we” to refer to the following groups of people?
- I am one with my [family/friends].
- I have a deep emotional bond with my [family/friends].
- [overlapping circles measures with family/friends]

Unity with distant Americans We measured unity with distant Americans using the same items adapted from the Identification with all Humanity scale and the pictorial identity fusion measures that were used to measure unity with close others. The distant Americans unity factor had a Cronbach’s alpha of 0.93.

Partisan animosity To assess partisan animosity, we measured both how open participants were to engaging with political opponents, and how warmly participants felt towards political out-group members. To measure openness to engagement, participants rated their openness to engaging with a political opponent on a 5-item Likert scale. To measure warmth towards political out-group members, participants rated their warmth towards a political out-group member on a feeling thermometer from 1 (very cold) to 10 (very warm).

Robustness measures

Political affiliation We asked participants to rate their political affiliation on a scale from “1: Extremely liberal” to “9: Extremely conservative.” From this measure, we also created a binary measure of participants’ strength of party affiliation in which participants who rated their political affiliation between a 3 and 7 were classified as politically moderate. Additionally, we asked participants to select whether the Democratic party or Republican party better represented their political views.

Local population demographics We used zip code census data to measure participants’ local population size, population density, racial diversity, and economic inequality. As a measure of local political composition, we also calculated the ratio of Democrats to Republicans in participants’ local communities based on voting data from the 2020 presidential election.

Physical distance from close others We asked participants to rate how closely they lived to each of the seven people interpersonally closest to them on a 4-pt ordinal scale from “We live together in the same home” to “We live several hours away from each other.”

Social network size We asked participants how many people they feel close to as a measure of total close other network size.

In-group warmth In addition to assessing warmth towards a political out-group member, participants rated their warmth towards a political in-group member on a feeling thermometer from 1 (very cold) to 10 (very warm). We also used this score to calculate an additional measure of warmth: an in-party minus out-party warmth difference score.

Network heterogeneity We asked participants to report the political affiliation and race of each of the seven people in their close-other social network. Using these responses, we calculated two measures of network heterogeneity. Political heterogeneity was operationalized as the average absolute difference between the participant's political affiliation and that of their close others. Racial diversity was calculated as the number of close others who shared the participant's racial identity.

Analytic strategy

In the structural equation model (SEM) depicted in Fig. 2, we tested the proposed model from density to partisan animosity. This SEM fit the data adequately, $\chi^2(338) = 1249.19$, $p < 0.001$, RSMEA = 0.071, SRMR = 0.089, CFI = 0.93, TLI = 0.92.

Results

Overall model

As predicted, analyses supported the overall model. Participants with denser networks reported feeling greater unity with close others (direct effect on unity with family: $\beta = 0.44$, $z = 10.48$, $p < 0.001$; direct effect on unity with friends: $\beta = 0.13$, $z = 2.85$, $p = 0.004$). Supporting the proposed mechanism of projected unity, close unity with family led to greater feelings of unity with distant Americans (direct effect: $\beta = 0.26$, $z = 5.24$, $p < 0.001$; indirect effect of network density through unity with family: $\beta = 0.11$, $z = 4.78$, $p < 0.001$), as did close unity with friends (direct effect: $\beta = 0.25$, $z = 5.61$, $p < 0.001$; indirect effect of network density through unity with friends: $\beta = 0.03$, $z = 2.56$, $p = 0.01$). In turn, participants who felt greater national unity reported more openness towards

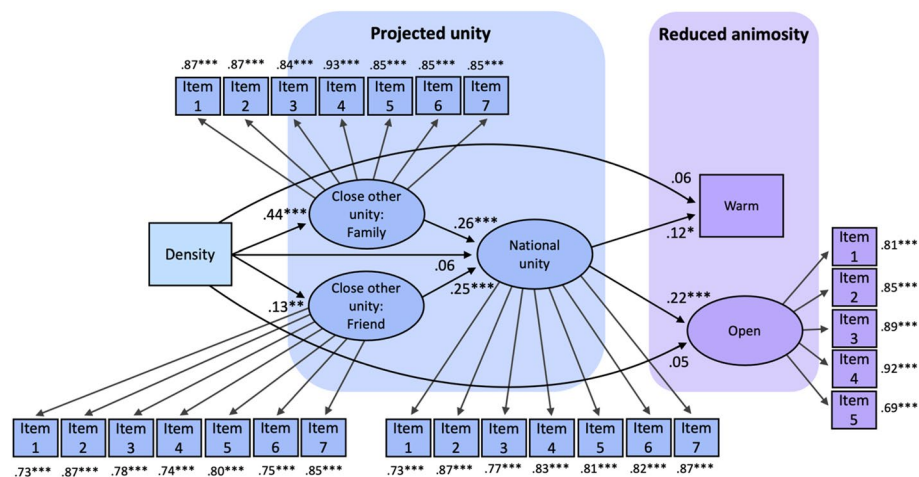


Fig. 2 Structural equation model results from Study 1. The model depicts the relationships between social network density, close other unity with family and friends, national unity with distant Americans, openness towards political opponents, and warmth towards political opponents. Arrows show direct effects. Indirect and total effects are reported in the text. Statistics are standardized betas. Single asterisks indicate significant p values of .05 or lower, double asterisks indicate p values of .01 or lower, and triple asterisks indicate p values of .001 or lower

political opponents (direct effect: $\beta = 0.22$, $z = 4.64$, $p < 0.001$; total indirect effect of close unity through unity with distant Americans: $\beta = 0.11$, $z = 4.02$, $p < 0.001$), and warmth towards political opponents (direct effect: $\beta = 0.12$, $z = 2.53$, $p = 0.01$; total indirect effect of close unity through unity with distant Americans: $\beta = 0.06$, $z = 2.41$, $p = 0.02$). This suggests that feelings of unity with close others were projected onto feelings of unity with distant Americans, which in turn correspond with lower partisan animosity. See Fig. 3 for relationships between social network density and dependent variables.

Mediation analyses

The mediational pathways from network density to partisan animosity supported the prediction that people with dense social networks feel a greater sense of unity with close others that is projected onto distant Americans, and these feelings of unity are associated with lower partisan animosity. The total indirect pathways from social network density to openness (total indirect effect: $\beta = 0.04$, $z = 3.27$, $p < 0.001$) and warmth (total indirect effect: $\beta = 0.02$, $z = 2.22$, $p = 0.03$) were significant, indicating that social network density indirectly impacted partisan animosity by fostering projected unity. When factoring in all pathways from network density to partisan animosity, social network density was associated with greater openness to political opponents (total effect: $\beta = 0.10$, $z = 2.34$, $p = 0.02$). However, although network density indirectly influenced warmth

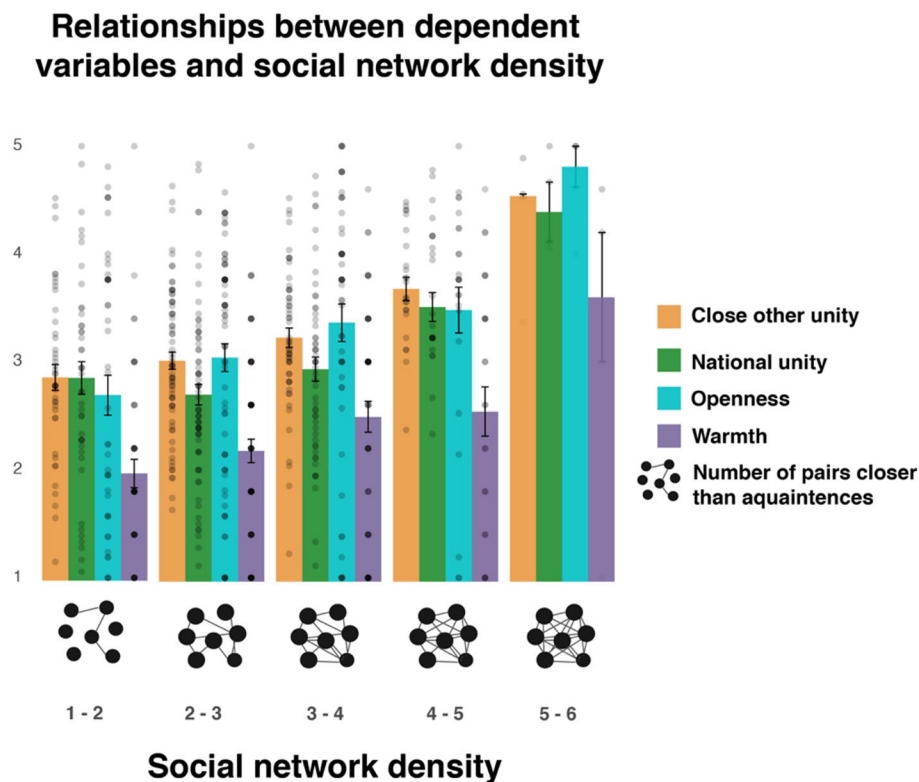


Fig. 3 Relationships between social network density and dependent variables in Study 1. The graph depicts the distribution of mean values of close unity, unity with distant Americans, openness towards political opponents, and warmth towards political opponents across density ranges in intervals from one (low social network density) to six (high social network density). Dependent variables are rescaled to a 1 (low) to 5 (high) scale. The error bars indicate standard errors. The networks represent the relationships between the seven people closest to the participants, and show the average number of people who are closer than acquaintances within each density range

through projected unity, the total effect of network density on warmth was not significant (total effect: $\beta = 0.07$, $z = 1.65$, $p = 0.10$).

We also examined the direct effects of network density on partisan animosity after accounting for the mediational pathways of projected unity. Projected unity fully mediated the relationships between density and partisan animosity such that the direct effects of social network density on openness ($\beta = 0.06$, $z = 1.36$, $p = 0.18$) and warmth ($\beta = 0.05$, $z = 1.09$, $p = 0.28$) were not significant. This suggests that projected unity fully explains the relationship between social network density and openness to political opponents. These findings suggest that people in dense social networks project feelings of unity with their close others onto feelings of unity with distant Americans, and that projected unity leads them to feel greater openness and, indirectly, greater warmth towards political opponents.

Robustness checks

Social network density with six-person networks We asked participants to report their close-other social network size, and found that the mean number of people participants felt close to was six people. Because our measure of social network density included seven people, it is possible that the social network density measure did not accurately reflect the density of many participants' proximate social networks because they were forced to add additional people with whom they did not feel as interpersonally close. We therefore investigated the relationships between partisan animosity and the social network density among the first *six* people participants listed as close others, reflecting the mean number of people participants reported feeling close with. We found that this six-person measure of social network density was significantly associated with warmth ($\beta = 0.09$, $t = 2.03$, $p = 0.04$) and openness ($\beta = 0.11$, $t = 2.56$, $p = 0.01$) towards political opponents. This further supported our core prediction that people with dense social networks tend to feel less partisan animosity.

In-party warmth Participants with denser social networks tended to report warmer feelings toward their own political party ($\beta = 0.09$, $t = 2.03$, $p = 0.04$), suggesting that social network density can foster not only openness to out-groups but also greater warmth towards one's in-group. This finding is consistent with the theorized mechanism of projected unity, in which individuals in denser networks generalize the cohesion they feel within their close-knit circles to broader identity groups, including political in-groups. Consistent with the main results, when both in-group and out-group warmth were included in the same model, out-group warmth was still not significantly associated with density in the full seven-person network version ($p = 0.087$). However, out-group warmth remained significant in the six-person social network analysis ($\beta = 0.09$, $t = 2.09$, $p = 0.037$), which is likely a more ecologically valid social network measure given that participants reported an average of six close ties.

The difference score between in-group and out-group warmth was not significantly associated with network density ($\beta = 0.003$, $t = 0.06$, $p = 0.95$). However, research on inter-group biases demonstrates that in-group sentiment is often independent of negative attitudes toward out-groups (Brewer 1999), and in the present study, we similarly found that in-group warmth and out-group warmth were largely uncorrelated ($r = 0.04$). Relatedly, past studies have demonstrated the separability of negative and positive partisanship (Bankert 2022, 2021). Given this pattern, it is arguably more theoretically valid to rely on

direct measures of animosity toward political out-groups rather than difference scores. Notably, our findings indicate that individuals embedded in dense social networks tend to express greater warmth toward both political in-groups and out-groups. This pattern is encouraging as it suggests that social network density may be associated with a broader openness to political engagement.

Political affiliation We tested whether the relationship between social network density and openness towards political out-group members remained after accounting for political affiliation. Social network density remained a significant predictor of openness ($\beta = 0.10$, $t = 2.25$, $p = 0.02$) after controlling for political affiliation and strength of political affiliation. This suggests that the association between dense networks and openness is robust across political ideologies. However, density remained a non-significant predictor of warmth (as it was in a linear model as a lone predictor: $\beta = 0.07$, $t = 1.64$, $p = 0.10$) after controlling for political affiliation and strength of political affiliation ($\beta = 0.06$, $t = 1.44$, $p = 0.15$). While social network density was not directly associated with warmth, its indirect effects through projected unity were significant (see main results).

Network heterogeneity We tested whether the political and racial heterogeneity of social networks moderated the relationship between network density and partisan animosity. We measured social network racial diversity as the number of close others who were of a different ethnicity than the participant, and social network political heterogeneity as the difference between the political leanings of the close others from that of the participant. We did not find a significant interaction between density and racial diversity on warmth towards political opponents ($\beta = -0.02$, $t(534) = -0.38$, $p = 0.70$), or openness towards political opponents ($\beta = -0.05$, $t(534) = -1.15$, $p = 0.25$). Similarly, we did not find a significant interaction between density and political heterogeneity on warmth towards political opponents ($\beta = 0.05$, $t(534) = 1.07$, $p = 0.29$), or openness towards political opponents ($\beta = 0.04$, $t(534) = 0.97$, $p = 0.33$).

Local population demographics We next examined the potential impacts of local demographic factors on the relationship between social network density and partisan animosity. First, it is possible that participants in smaller towns have denser social networks, and that the impact of network density on partisan animosity is simply due to differences in animosity between cities and towns. To test this possibility, we added population size and population density as control variables in linear regressions with density and partisan animosity. Controlling for population size and population density, social network density remained a significant predictor of openness ($\beta = 0.12$, $t = 2.66$, $p = 0.008$), but remained a non-significant predictor of warmth ($\beta = 0.08$, $t = 1.72$, $p = 0.09$).

Additionally, regional racial diversity, political demographics, and economic inequality are factors that could impact group relations and potentially increase partisan animosity. To test the robustness of our results against regional factors, we added political composition, racial diversity, economic inequality, population size, and population density as control variables in linear regressions of density and partisan animosity. Controlling for these variables, social network density remained a significant predictor of openness ($\beta = 0.12$, $t = 2.61$, $p = 0.009$), but remained a non-significant predictor of warmth ($\beta = 0.08$, $t = 1.78$, $p = 0.08$). While social network density was not directly associated with warmth, its indirect effects through projected unity were significant (see main results).

Other indicators of strong social connections Exploratory analyses further supported the idea that people with more robust social connections show less partisan animosity. Participants living closer to their seven closest acquaintances showed more openness ($\beta = 0.13$, $t(490) = 2.97$, $p = 0.003$) and warmth ($\beta = 0.10$, $t(490) = 2.12$, $p = 0.03$) to political adversaries. Additionally, participants with a greater number of close others felt more openness towards political opponents ($\beta = 0.18$, $t(536) = 4.32$, $p < 0.001$), but there was no significant relationship between the number of close social ties and warmth ($\beta = 0.07$, $t(536) = 1.55$, $p = 0.12$).

Discussion

As predicted, participants with denser social networks felt a heightened sense of unity with their close connections which was projected onto broader national feelings of unity, encompassing distant Americans. Participants who showed this projected unity felt more warmth towards political opponents, and openness towards a “political opponent” they believed they would be meeting. Social network density was directly associated with greater openness towards political opponents across political ideologies, among both political moderates and strongly identified partisans, and after factoring in regional demographics including local political composition, population size, racial diversity, and economic inequality. These results suggest that people in dense social networks may often feel lower partisan animosity.

Study 2: manipulating density

Having found evidence that people in denser social networks show greater openness towards political opponents, and, indirectly, greater warmth via projected unity, we next explored a potential intervention for reducing partisan animosity: encouraging people to focus on the relationships among densely connected people in their social networks. We found in Study 1 that people who *perceived* their social network to be denser reported greater openness to political opponents (see Supplementary Materials). While it is challenging to experimentally manipulate social networks, social connection is often based more on perception than objective indicators (Haber et al. 2007; Dias et al. 2018; Mund et al. 2020). Accordingly, we hoped that it may be possible to lower partisan animosity—via the paths outlined in the model (Fig. 1)—without changing the actual composition of people’s social networks, by leading them to focus on the parts of their social networks that are dense. However, it is also important to acknowledge that impacting people’s psychological experiences enough to reduce partisan animosity may require actual structural changes in individuals’ social networks or the dynamics within them—changes that are difficult to influence outside of natural experiments or sustained, real-life interventions.

With these potential benefits and challenges in mind, we developed an intervention aimed at directing participants’ attention to densely connected parts of their social networks. Using a between-subjects design, participants in the “dense condition” listed four close others who knew each other well, while those in the “sparse condition” listed four close others who did not know each other well. We hypothesized that perceiving greater social network density would cause a cascading effect whereby participants focusing on densely connected relationships would report stronger unity with their close friends and

family, which, in turn, would extend to a greater sense of unity with distant others, ultimately reducing feelings of partisan animosity.

Methods

Preregistration

This study is preregistered at <https://osf.io/fpz2v/>.

Participants

We recruited 481 participants ($M_{\text{age}} = 40.16$, $SD_{\text{age}} = 11.37$) from MTurk. Of these, 32 did not pass an attention check and/or manipulation check, leaving a remaining 449 participants. 222 of the participants were men, 221 were women, 2 were non-binary, and 4 did not specify their gender. 7.3% were Hispanic or Latino/a/x, 68.8% were White, 8.9% were Black, 9.8% were Asian, 0.68% were Native American or Alaskan native, 3.3% were multiracial, and 1.1% did not specify their race.

Procedure

Participants were assigned to one of two conditions. In a manipulation of perceived network density adapted from (Lee et al. 2020), participants either listed four people they feel close with who know each other well, or four people they feel close with who do not know each other well. These were the instructions in each condition:

*“From time to time, people receive support from different individuals **who (also/ don’t) know each other well**. Please think of four people in your life—who are (not) close to one another—that you go to for support.”*

Participants in the dense network condition reported four relationships from their social network that were more densely connected than did participants in the sparse condition, indicating that the manipulation was successful, $\beta = 1.32$, $t(465) = 19.04$, $p < 0.001$. Consistent with prior work using this method (Lee et al. 2020), we did not include a traditional control group (e.g., no network reflection), as both conditions ask participants to consider real aspects of their social networks. This approach helped ensure ecological relevance and allowed us to directly compare the effects of dense versus sparse perceptions of social connections—our primary theoretical contrast.

Next, we assessed partisan animosity using the same methods as in Study 1. Participants then completed the unity measures, described in detail in Study 1.

Main measures

Focused social network density We first asked participants to list four people interpersonally close to them who either knew each other well (dense condition) or did not (sparse condition). We then asked them to report the interpersonal closeness among these four people on a 6-pt scale ranging from complete strangers to extremely close. Density was calculated as the sum of the strengths of the relationships between each pair in the network divided by the total number of pairs.

We asked participants to report four people instead of seven (as they did in Study 1) because participants with particularly dense proximate social networks may not have many close others who do not know each other, and conversely, participants with

particularly sparse proximate social networks may not have many close others who do know each other.

Other measures We measured unity with close others, unity with distant Americans, and partisan animosity in an identical manner to Study 1.

Model

In the structural equation model (SEM) depicted in Fig. 4, we tested a model in which focusing on the density (vs. sparseness) of one's own network predicted unity with close others, that unity with close others predicted unity with distant Americans, and this unity with distant Americans predicted warmth and openness towards political opponents. The structural equation model allowed us to assess the fit of this proposed model and to test for the indirect effects of unity on the relationships between density and partisan animosity. The model fit the data adequately, $X^2(338) = 1344.68$, $p < 0.001$, RSMEA = 0.076, SRMR = 0.11, CFI = 0.92, TLI = 0.91.

Results

Effects of manipulation

Social networks are challenging to experimentally manipulate, but we hoped that the “focusing on density/sparseness” manipulation might directly impact partisan animosity. The manipulation did not significantly affect warmth (total effect: $\beta = 0.03$, $z = 0.69$, $p = 0.49$) or openness (total effect: $\beta = 0.03$, $z = 0.73$, $p = 0.47$) towards political opponents. However, consistent with the proposed model, the manipulation indirectly influenced partisan animosity through a series of indirect effects.

Full model: manipulation to projected unity

As predicted, participants in the dense experimental condition reported denser “focused” social networks than participants in the sparse condition ($\beta = 0.68$, $z = 19.74$, $p < 0.001$). In turn, participants who reported denser focused networks reported feeling

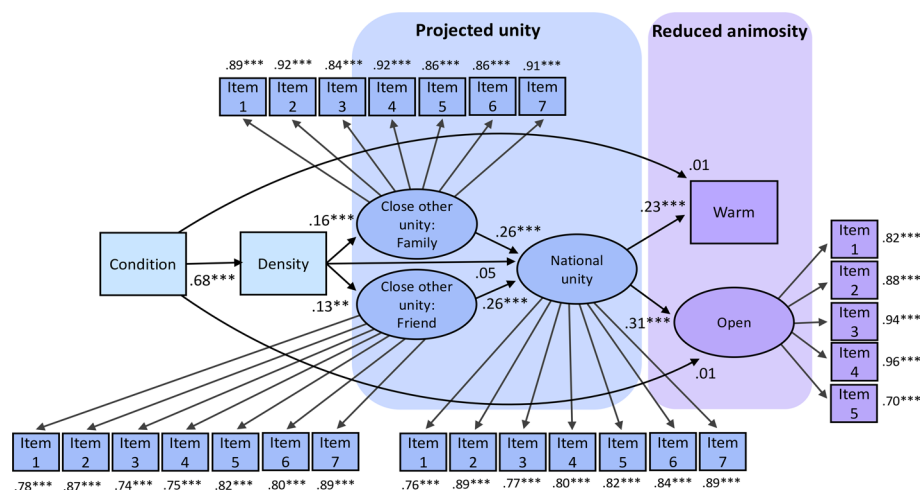


Fig. 4 Structural equation model of results from Study 2. The model depicts the relationships between the experimental condition, focused social network density, close other unity with family and friends, national unity, openness towards political opponents, and warmth towards political opponents. Arrows show direct effects. Indirect and total effects are reported in the text. Statistics are standardized betas. Single asterisks indicate significant p value of .05 or lower, double asterisks indicate p value of .01 or lower, and triple asterisks indicate p value of .001 or lower

greater unity with family (direct effect: $\beta = 0.16$, $z = 3.35$, $p = 0.001$; indirect effect through experimental condition: $\beta = 0.11$, $z = 3.30$, $p = 0.001$) and friends (direct effect: $\beta = 0.13$, $z = 2.56$, $p = 0.01$; indirect effect through experimental condition: $\beta = 0.09$, $z = 2.54$, $p = 0.01$). Supporting the proposed mechanism of projected unity, close unity with family led to greater feelings of unity with distant Americans (direct effect: $\beta = 0.26$, $z = 5.30$, $p < 0.001$; indirect effect of network density through unity with family: $\beta = 0.04$, $z = 2.86$, $p = 0.004$), as did close unity with friends (direct effect: $\beta = 0.26$, $z = 5.26$, $p < 0.001$; indirect effect of network density through unity with friends: $\beta = 0.03$, $z = 2.31$, $p = 0.02$).

Full model: projected unity to partisan animosity

Participants who felt greater unity with distant Americans reported more openness towards political opponents (direct effect: $\beta = 0.31$, $z = 6.07$, $p < 0.001$; total indirect effect of close unity through unity with distant Americans: $\beta = 0.16$, $z = 4.82$, $p < 0.001$), and warmth towards political opponents (direct effect: $\beta = 0.23$, $z = 4.66$, $p < 0.001$; total indirect effect of close unity through unity with distant Americans: $\beta = 0.12$, $z = 4.02$, $p < 0.001$). This suggests that feelings of unity with close others were projected onto feelings of unity with distant Americans, which in turn correspond with lower partisan animosity.

Indirect effects of manipulation on partisan animosity

The manipulation was indirectly associated with increased warmth and openness by increasing focused social network density and projected unity. The total indirect pathways from the experimental condition to openness (total indirect effect: $\beta = 0.03$, $z = 2.30$, $p = 0.02$) and warmth (total indirect effect: $\beta = 0.02$, $z = 2.19$, $p = 0.03$) were significant, indicating that the manipulation indirectly impacted partisan animosity.

Indirect effects of focused network density on partisan animosity

The mediational pathways from focused social network density to partisan animosity supported the prediction that dense social networks can lower partisan animosity through feelings of unity with close others that are projected onto distant Americans. The total indirect pathways from focused social network density to openness (total indirect effect: $\beta = 0.02$, $z = 3.17$, $p = 0.002$) and warmth (total indirect effect: $\beta = 0.02$, $z = 2.90$, $p = 0.004$) were significant, indicating that social network density indirectly impacted partisan animosity by fostering projected unity.

Discussion

The results provide additional support for our model linking social networks to partisan animosity while also highlighting the inherent challenges of directly manipulating social network features. Although the quick framing intervention did not significantly reduce partisan animosity, it nonetheless produced a causal cascade in the expected directions. Specifically, the manipulation increased focused network density, which was associated with greater unity with close others and, in turn, with distant Americans, and was ultimately indirectly linked to lower partisan animosity. However, the total effects of the manipulation on partisan animosity were not significant.

Experimentally manipulating social network density is difficult, particularly through a short-term intervention like the one used here which influences focused rather than

actual network density. Social networks are deeply ingrained and shaped by long-term interactions and experiences, and even when perceptions are nudged, this does not necessarily replicate the sense of community that emerges from truly dense social networks. Another limitation of our approach is that participants may not have had four individuals in their lives who precisely fit the intended criteria—those they turn to for support but who either do or do not know each other well. We also acknowledge the lack of a traditional “no-intervention” control as a limitation. Following prior work (Lee et al. 2020), we compared two ecologically valid conditions involving social network reflection to isolate the effect of perceived density. However, without a true control, we cannot rule out that general aspects of the task influenced the results.

Ultimately, impacting partisan animosity through network density may require long-term interventions that change the actual composition or dynamics of people’s social networks. Given the preliminary evidence partially supporting the model’s pathways through a series of indirect effects, future studies could explore more sustained methods of fostering dense social connections.

Study 3: community social network density

The results of Studies 1 and 2 suggest that people with denser social networks may exhibit lower partisan animosity. But does this relationship emerge at the community level across the United States? Here, we investigated whether denser community social networks exhibit lower partisan animosity.

To measure social network density, we used the community cohesiveness metric from the Social Capital Atlas, which captures the likelihood that two Facebook friends in a given zip code or county are also friends with each other. We conducted two analyses using this metric. First, we examined our own participants—1,863 individuals from 1,602 postal codes across each study we conducted in the present work that included measures of warmth and openness toward political opponents—to test whether those living in areas with denser social networks reported lower levels of partisan animosity. Second, we ran a regional-level analysis using an explicit attitudes measure of partisan animosity from a Project Implicit dataset, which provided scores of warmth towards political opponents for 79,819 participants across 154 counties. In both cases, we used the Social Capital Atlas data to determine the social network density of each zip code or county. These two approaches allowed us to assess the relationship between community network density and partisan animosity at both the individual and regional levels.

Methods

Preregistration

This participant-level analysis of community social network density and partisan animosity is preregistered as an exploratory prediction for Studies 1 and 2 at <https://osf.io/fpz2v/>.

Procedure

We examined the relationship between community social network density and partisan animosity using two complementary approaches. In both cases, we used the community cohesiveness measure from the Social Capital Atlas, developed by Opportunity Insights at Harvard University. This measure, based on aggregated analyses of 21 billion

Facebook friendships, estimates the likelihood that two friends of a person in a given postal code or county are also friends with each other. While Facebook friendships do not perfectly capture social ties—given that not all individuals have Facebook accounts and that online connections do not necessarily reflect relational closeness—this measure offers a valuable approximation due to its extensive coverage, encompassing billions of friendships across the United States. There is also strong precedent for using Facebook-based network measures in the literature, where they have been shown to meaningfully capture local social structure and predict a range of social and political outcomes (Lewis et al. 2008; Mayer and Puller 2008).

Participant-level analysis In our first analysis, we looked at whether individuals living in areas with higher social network density reported lower levels of partisan animosity. We combined data from Studies 1, 2, S1, S2, and S3, resulting in a total sample of 1,863 participants from 1,602 postal codes. Each participant's community cohesiveness score was matched to their postal code using the Social Capital Atlas data. We then tested whether those in denser social environments showed more warmth and openness toward political opponents.

Regional-level analysis Our second analysis examined partisan animosity at the regional (county) level using an explicit attitudes measure from a publicly available Project Implicit dataset. This dataset includes scores of warmth toward political opponents, averaged at the county level, from a total of 79,819 participants across 154 counties. We matched these counties to their corresponding Social Capital Atlas cohesiveness scores to test whether regions with denser social networks tend to exhibit lower levels of partisan animosity.

Main measures

Social network density To measure community social network density, we used the county cohesiveness measure from the Social Capital Atlas. For each Facebook user, cohesiveness is defined as the proportion of their friends who are also friends with one another. These individual cohesiveness scores are then averaged at the zip code or county level to produce a community-level measure. The data for this measure was collected in 2019.

Partisan animosity To assess partisan animosity, we used 1) Combined postal code level data from all of the present studies that assessed participant-level partisan animosity (Studies 1, 2, S1, S2, and S3) on how open participants were to engaging with political opponents, and how warmly participants felt towards political out-group members 2) County-level data on warmth towards political opponents from Project Implicit as a regional measure of partisan animosity. We created a measure of partisan animosity by examining the average level of warmth towards political opponents per county, using the 2019 wave of data from the Political Identity IAT to match the year of the cohesiveness measure used to assess community social network density. Participants of the Political Identity IAT responded to the question “How warm or cold do you feel towards the following political party?” for both Democrats and Republicans on a scale from “0: Extremely cold” to “10: Extremely warm.” We examined Democrat participant warmth towards Republicans, and Republican participant scores towards Democrats for all counties with at least ten participant responses.

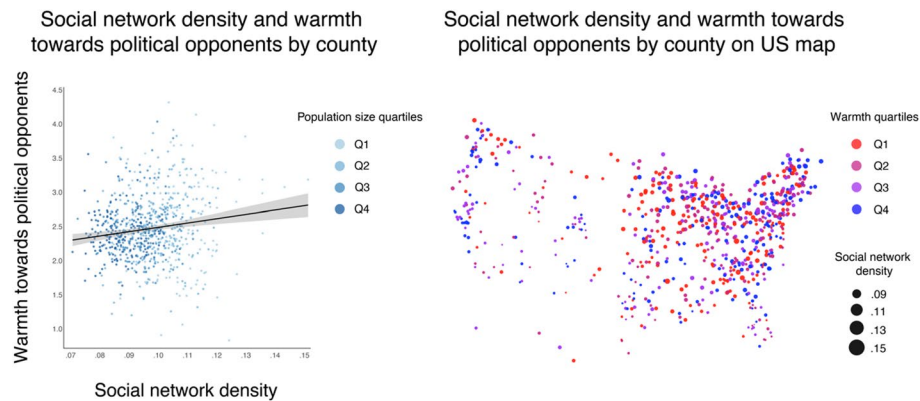


Fig. 5 The relationship between social network density and warmth towards political opponents by county, as measured from Project Implicit aggregated county scores. Warmer colors represent counties with higher average warmth towards political opponents. Left: Line plot of social network density and warmth towards political opponents by county. Color indicates population size, whereby darker data points represent counties with larger populations. Shaded area surrounding line represents standard error. Right: Counties are depicted on a United States map. Dot size represents social network density whereby larger dots represent counties with denser social networks

Results

Participant-level partisan animosity

As predicted, participants living in postal codes with higher social network density showed lower partisan animosity (warmth: $\beta = 0.06$, $t(1855) = 2.63$, $p = 0.009$; openness: $\beta = 0.07$, $t(1855) = 2.98$, $p = 0.003$). Additionally, when controlling for population size, regional political composition, racial diversity, and economic inequality, community social network density was still significantly associated with warmth ($\beta = 0.09$, $t(1843) = 2.94$, $p = 0.003$) and openness ($\beta = 0.08$, $t(1845) = 2.70$, $p = 0.007$). These results suggest that people living in densely interconnected communities show less partisan animosity.

Regional partisan animosity

As predicted, counties with higher community social network density had lower rates of partisan animosity, $\beta = 0.11$, $t(876) = 3.23$, $p = 0.001$. Given that small towns are generally denser than larger cities, we next controlled for county population size to see whether this relationship remained significant regardless of county population. The positive relationship between social network density and warmth remained after controlling for population size, $\beta = 0.09$, $t(876) = 2.47$, $p = 0.01$. This suggests that communities with more interconnected social networks exhibit less partisan animosity. See Fig. 5 for relationship between social network density and warmth towards political opponents by county.

Perhaps, counties with higher social network density are only warmer towards political opponents in places that are not highly politically split, or in places with certain political demographics. We tested the extent to which the relationship between social network density and warmth generalized after controlling for regional political demographics. Social network density remained significantly associated with warmth after controlling for both the extent of political variation (how evenly divided the compositional split between the two parties is, $\beta = 0.11$, $t(875) = 3.27$, $p = 0.001$), and the relative percentage of Democrats versus Republicans ($\beta = 0.08$, $t(875) = 2.49$, $p = 0.01$). Community

social network density appears to be associated with warmth regardless of local political demographics.

Counties with higher average social network density also reported significantly greater warmth toward political in-groups ($\beta = 0.13$, $t = 4.00$, $p < 0.001$) in addition to out-groups ($\beta = 0.16$, $t = 4.66$, $p < 0.001$), even when both predictors were included in the same model. This suggests that denser community networks are associated not only with reduced partisan animosity but with elevated political warmth more broadly. The warmth difference score between in-group and out-group warmth was unrelated to social network density ($\beta = 0.01$, $t = 0.23$, $p = 0.82$). However, as prior research on intergroup bias show that feelings toward political out-groups are often independent of feelings toward one's political in-group (Bankert 2022, 2021), using direct measures of out-group animosity may provide a more conceptually accurate approach than relying on difference scores. These findings replicate the Study 1 in-group warmth findings and indicate that denser regions may foster broader political openness.

To further test the robustness of the relationship between community social network density and political out-group warmth, we simultaneously controlled for several regional factors that could impact partisan dynamics. After controlling for population size, regional political composition, racial diversity, and economic inequality, community social network density was still significantly associated with lower partisan animosity, $\beta = 0.09$, $t(871) = 2.34$, $p = 0.02$. These results suggest that densely interconnected communities have less partisan animosity than sparsely connected communities across a wide range of regional factors.

Regional partisan animosity: replication

We replicated our regional partisan animosity findings using a political outgroup warmth measure from (Lelkes et al. 2017). This measure is based on feeling thermometer ratings toward political outgroup candidates, originally collected in the 2004 and 2008 waves of the National Annenberg Election Studies, a nationally representative survey of U.S. voters (139,389 participants across 3,032 counties). Replicating our analysis while controlling for county population size, we found that counties with denser community social networks exhibited lower levels of partisan animosity, $\beta = 0.05$, $t(2905) = 2.41$, $p = 0.01$. Using a difference score (ingroup minus outgroup warmth), we again observed reduced animosity in denser counties, $\beta = -0.06$, $t(2903) = -3.09$, $p = 0.002$. These results provide convergent evidence from an additional dataset.

Discussion

We found that community social network density predicted lower partisan animosity both among individual participants across the studies in the present work, and counties across the United States. In the participant-level analysis, individuals residing in zip codes with denser social networks reported greater warmth and openness toward political opponents.

In the regional analysis, counties with higher average community social network density displayed greater warmth towards political opponents. The positive associations of community social network density with participant- and county-level partisan animosity persisted even when accounting for potential confounding variables like population size, regional political composition, racial diversity, and economic inequality. Moreover,

we replicated the main finding using an additional dataset of partisan animosity. These results suggest that communities with denser social connections tend to exhibit less divisive attitudes towards political opponents.

General discussion

Partisan animosity and social isolation in the United States have both escalated in recent years (Boxell et al. 2022; Hampton et al. 2009; Holt-Lunstad et al. 2010; Wilson and Moulton 2010). Americans interact with others less in general (Holt-Lunstad et al. 2010; Wilson and Moulton 2010), and avoid those who belong to a different political party (Gimpel and Hui 2015). While many argue that the nation needs more interconnected communities to mitigate loneliness (Holt-Lunstad et al. 2017), our findings suggest that denser social networks may also help bridge political divides.

We found that people embedded in dense social networks felt an elevated sense of unity with their close others which they extended to distant Americans. This sense of national unity translated into higher openness to engaging with political opponents. Similarly, communities with denser networks held greater warmth and openness towards political opponents. Participants located in regions with denser community networks exhibited greater warmth and openness to political opponents compared to those living in less interconnected regions, even after accounting for regional factors like population size, political demographics, racial diversity, and economic inequality, a range of factors which can impact group dynamics (Bradley and Chauchard 2022; Gidron et al. 2020; Martin and Webster 2020; Rogowski and Sutherland 2016).

Dense community networks were associated with lower partisan animosity regardless of the region's political climate—be it one that is highly politically polarized or one where political views are largely homogenous, which is noteworthy because people living in regions with limited exposure to political opponents tend to feel more partisan animosity (Smiley 2020). Exploratory analyses suggested that people who lived in closer physical proximity to their closest social relationships also showed more warmth and openness towards political opponents, lending further credence to the idea that unified social connections can mitigate political hostilities.

The relationship between social network density and openness towards political opponents was more robust than its relationship with warmth towards political opponents. While *community* social network density robustly predicted warmth at both the participant and regional levels (with the regional effect replicated in an additional dataset), the social network density of individual participants did not. However, an analysis using individual networks composed of the average number of close connections participants reported (six, one fewer than the seven total measured) did show an association with greater warmth toward political opponents. Nonetheless, the evidence for an association between individual network density and warmth remains mixed and warrants further investigation.

It stands to reason that being part of a dense network may encourage openness towards political opponents more than warmth. The interconnectedness of dense networks necessitates openness to resolving conflicts—even with those one may not feel warmly towards—to maintain group harmony. Openness to discussion is particularly important for reducing partisan animosity because cross-party conversations have been found to increase positive regard towards political opponents (Levendusky and Stecula

2021). Given that partisans in today's political climate are generally opposed to interacting with political opponents (Gimpel and Hui 2015), a mere willingness to engage in conversation is a crucial step towards mending political rifts.

We also acknowledge that while social network density was connected to warmth toward out-group members, it did not significantly predict the typically used difference score of out-group-minus-in-group warmth. On one hand, it makes sense to control for how warm people generally feel towards others. On the other hand, the direct warmth measure used in the present work may better capture out-group animosity (Kubin and Sikorski 2024), given evidence that such animosity is often independent of in-group sentiment (Brewer 1999), and that negative partisanship is distinct from positive partisanship (Bankert 2022, 2021). It is encouraging that social network density was also positively associated with political *in-group* warmth at both individual and regional levels, aligning with the projected unity mechanism. At a time when many Americans feel disillusioned with politics—including their own party (Pew Research Center. Americans' dismal views of the nation's politics 2023)—these findings suggest that strengthening community ties may broadly foster political openness.

More broadly, we caution that our main findings are correlational and encourage future research to test for a causal relationship between social network density and partisan animosity. To explore potential applications of these insights, we tested an intervention that encouraged participants to focus on denser parts of their social networks. However, the manipulation did not directly reduce partisan animosity, possibly because it did not alter the actual composition of participants' networks. Although social networks are difficult to manipulate experimentally (Phan and Airolidi 2015; Brashears and Gladstone 2020), longer-term interventions that increase network density—such as brokering introductions among friends and family or organizing community activities—are worth testing for their potential to reduce partisan animosity.

Achieving network-related reductions in partisan animosity may require long-term, sustained increases in social density to foster the sense of unity that emerges from more interconnected relationships. In line with several intervention studies in medicine (Stull et al. 2022), biology (Warner and Lovern 2014), family intervention (Mason et al. 2016; Pasalich et al. 2016), psychology (Buehler and High 2023; Chen and Attrition 2013; Kirkpatrick and Locke 1996), and behavioral therapy (Carpenter et al. 2019), we examined the indirect effects of the manipulation and found that it was indirectly associated with lowered partisan animosity through its effects on focused social network density and projected feelings of unity—a “causal cascade” (Stull et al. 2022). These findings suggest a promising direction for future, more intensive interventions.

The present findings offer novel insights into mitigating partisan hostility by highlighting the role of social network structures. Building on previous research that has identified key interpersonal factors that shape partisan animosity—including partisan sorting (Mason 2015, 2018), interparty contact (Wojcieszak and Warner 2020), and misperceptions of political opponents (Enders and Armaly 2019; Lees and Cikara 2020; Moore-Berg et al. 2020)—our results underscore the importance of considering the broader architecture of individuals' social environments. Specifically, the perceived sense of unity within one's immediate social circle appears to influence perceptions of national cohesion. Prior work rooted in contact theory has shown that interparty interactions can reduce partisan hostility by fostering a shared identity between the self and the political

out-group (Wojcieszak and Warner 2020). Our findings that accounting for network political composition did not alter the relationships between social network density and partisan animosity suggest that reductions in animosity may emerge even in the absence of direct interparty contact. A cohesive and unified local social context—regardless of political composition—may generalize outward, promoting greater openness and reducing partisan hostility at the societal level.

The present studies demonstrate a novel potential mechanism for promoting positive relations between groups: projected unity. Coinciding with prior work demonstrating that people use their local environment to simulate the global environment (Dickie et al. 1997; Barsalou and Wiemer-Hastings 2005; Lakoff et al. 1999), participants with denser social networks consistently exhibited feelings of unity not just with their immediate social circles, but extended to Americans at large. These results build on past work finding that people with more robust social support exhibit more prosocial behavior, suggesting that the benefits of close relationships can extend to generalized social concern (Pavey et al. 2011; Li et al. 2021; Guzman et al. 2012; Guo 2017; Lin et al. 2015). This insight has the potential to inform new bottom-up interventions focused on promoting unity within local social circles to help bridge larger societal divides: When people are embedded in close communities, they appear to project the feelings of cohesion they experience onto their perceptions of broader groups, perceiving political opponents less as out-group antagonists and more as fellow members of the nation they share.

We encourage future research to continue exploring social network characteristics to discern which kinds of dense groups are most effective at fostering projected unity and reduced animosity. Certain group characteristics may moderate the relationship between social network density and projected unity, like group diversity, norms, and insularity (Terry et al. 2004; Oliver and Wong 2003). We did not find interactions between social network density and social network political or racial diversity on animosity, but future work may uncover important factors we did not explore. Given the rich potential theoretical basis for the association between social network density and reduced animosity, future research could also investigate other plausible mediators—such as psychological security, meta-perceptions of others' partisan animosity, or approaches to conflict—that may co-vary with dense networks and shape intergroup attitudes.

Further examining projected unity as a potential mechanism to promote intergroup warmth could offer valuable insights. Future studies could explore the possibility of leveraging unity from close relationships to distant ones in efforts to mitigate racism, sexism, and other forms of discrimination. Promisingly, we found that people embedded in dense networks extended their feelings of unity to people around the globe (see Supplementary Materials). This suggests that social cohesion from dense networks may not be limited to political or national identities, but under the right conditions could reflect a broader, more inclusive perception of humanity. Investigating the link between social network density and xenophobia could reveal whether projected unity can amplify concern for humanity at large.

As the United States faces growing political divides, the present research proposes a novel potential solution of cultivating denser social networks. Dense social networks are tied to feelings of unity which, when projected at the national level, could possibly reduce partisan animosity and encourage constructive dialogue between political opponents. The possibility for dense networks to act as a unifying force warrants further

investigation, as closer-knit communities could be a key step toward a less divided nation.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s41109-025-00742-7>.

Supplementary Material 1

Author contributions

D.D., K.L., and K.G. conceived and designed the research. D.D. analyzed the data. D.D., K.L., and K.G. wrote the manuscript. D.D. and K.G. produced the figures and tables.

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Data availability

The data and code necessary to replicate this project are available at <https://osf.io/fpz2v/>.

Declarations

Competing interests

The authors declare no competing interests.

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